

MailShield AI: Plain-Language Reference Guide

All Key Terms, Features, Architecture & Real-World Examples Explained in Simple Words

Purpose of this Document: Cybersecurity and email forensics contain complex jargon (like RFC-5322, DKIM RSA, MTA unrolling, SHA-256 custody). This document translates **every term and every feature of MailShield AI** into plain, easy-to-understand English with **real-world analogies and everyday examples**, so anyone—from a beginner student to a non-technical judge—can instantly understand how the platform works.

CHAPTER 1: ALL CYBERSECURITY TERMS EXPLAINED SIMPLY

1. Email Header (The Invisible Envelope)

Simple Definition: The header is the hidden technical 'envelope' of an email. While you normally only see the body message, the header records every post office server that touched the email, the sender's real IP address, and security checkmarks.

Real-World Example: Think of a postal letter. The email body is the letter inside. The email header is the physical envelope stamped by post offices in Delhi, Dubai, and London, showing the exact date, time, and postmark of every stop.

2. RFC-5322 (The Internet Mail Rulebook)

Simple Definition: RFC-5322 is the official international specification that dictates how every legitimate email in the world must be formatted.

Real-World Example: Like the standard passport format worldwide. Every passport has a Photo, Country Code, and Expiry Date in the exact same order. RFC-5322 ensures every email has a valid 'From', 'Date', 'To', and 'Subject' format.

3. MTA — Mail Transfer Agent (The Digital Post Offices)

Simple Definition: An MTA is a specialized server computer that receives, routes, and delivers emails across the internet.

Real-World Example: When you mail a package from Kolkata to London, it goes from Kolkata Sorting Hub → Mumbai Airport Hub → Heathrow Sorting Office. Each of those sorting hubs is an MTA.

4. Transit Hop (The Flight Layovers)

Simple Definition: Each server an email passes through on its way from sender to receiver is called a 'hop'. Every hop appends a 'Received:' header line.

Real-World Example: If a flight flies Delhi → Doha → London, it took 2 hops. MailShield unrolls these hops like an airline boarding itinerary.

5. Originating IP (The Attacker's Physical Address)

Simple Definition: The unique numerical internet address (like 185.220.101.5) of the original computer or server that clicked 'Send'.

Real-World Example: The CCTV footage of the specific street mailbox where the criminal originally dropped the fake letter.

6. SPF — Sender Policy Framework (The Authorized Guest List)

Simple Definition: SPF allows a company (like Google or your bank) to publicly list all the official server IP addresses allowed to send emails using their domain name.

Real-World Example: A bank tells building security: 'Only delivery trucks with license plates ABC-123 and XYZ-789 are allowed to deliver mail for us. If any other truck arrives, reject it!' When a hacker sends an email from an unauthorized truck, SPF fails.

7. DKIM — DomainKeys Identified Mail (The Royal Wax Seal)

Simple Definition: DKIM attaches a mathematical cryptographic signature to the email. The recipient uses the sender's public key to verify that not a single word was altered in transit.

Real-World Example: A medieval king stamps his letter with a unique royal wax seal. If a bandit intercepts the letter and changes 'Pay 100 Gold' to 'Pay 10,000 Gold', the wax seal breaks and the recipient knows it was tampered with.

8. DMARC (The Chief Security Guard)

Simple Definition: DMARC brings SPF and DKIM together. It checks whether the name shown to the user in 'From:' matches the authorized SPF/DKIM domains, and tells receiving servers what to do with fakes (monitor, spam, or reject).

Real-World Example: The security guard checks both the delivery truck (SPF) and the wax seal (DKIM). If both fail or the driver wears a fake SBI badge, the guard burns the letter immediately (p=reject).

9. Return-Path vs Visible 'From:' (The Fake Sender ID)

Simple Definition: The 'From:' line is just text typed by the sender (easy to fake). The 'Return-Path' is the actual technical address where error bounces go.

Real-World Example: A scammer writes 'Police Commissioner' at the top of a letter (From:), but the return envelope address at the back points to criminal@darkweb.ru (Return-Path). MailShield flags this mismatch instantly.

10. SHA-256 Custody Hash (The Digital Fingerprint)

Simple Definition: A mathematical algorithm that turns any email into a unique 64-character code. Even if an attacker alters a single comma, the hash completely changes.

Real-World Example: DNA or fingerprint evidence sealed in a police tamper-proof plastic evidence bag. It proves to a judge in court that the evidence was not modified.

11. Chain of Custody & Section 65B (Court Admissibility)

Simple Definition: A legal requirement under Section 65B of the Indian Evidence Act stating that digital evidence is only admissible in court if you can prove who collected it, when, and that it was never altered.

Real-World Example: A police evidence locker where every officer who touches the weapon must sign the logbook. MailShield automatically creates this certificate for every scan.

12. ASN — Autonomous System Number (Network Owners)

Simple Definition: A global identifier for large internet networks. For example, AS15169 is Google LLC; AS49981 is WorldStream Netherlands.

Real-World Example: Like airline operator codes: AI for Air India, BA for British Airways. It tells us who actually owns the communication pipes.

13. Typosquatting & Lookalike Domains (The Clever Copycats)

Simple Definition: When criminals register domain names that look almost identical to legitimate brands to trick human eyes.

Real-World Example: Buying paypa1.com (with the number 1 instead of letter l) or icici-bank-kyc-update.top to trick users into entering net banking passwords.

14. Business Email Compromise — BEC (The CEO Scam)

Simple Definition: A targeted phishing attack where a criminal pretends to be the CEO, Director, or Vendor and emails the accountant asking for an urgent financial wire transfer.

Real-World Example: An email sent at 6 PM on Friday pretending to be your Boss: 'I am in an emergency meeting. Transfer Rs. 5 Lakhs to this vendor account right now.'

CHAPTER 2: ALL WEBSITE FEATURES EXPLAINED WITH EXAMPLES

1. Multi-Input Email Scanner (/scan)

What It Does: The main investigation portal where you can either drag-and-drop a saved .eml email file or paste raw header text copied directly from Gmail or Outlook.

How to Use: *Open Gmail on your computer, click 'Show original' → 'Copy to clipboard', paste it into MailShield, and click 'Analyze Now'.*

2. 1-Click Demo Phishing & Safe Sample Buttons

What It Does: Built-in sample buttons that instantly load authentic, pre-configured phishing and safe email headers with one click so you never have to search for sample files during demos.

How to Use: *On the scan page, click 'Try Sample Phishing Email' to instantly load a real Russian MTA credential-harvesting phishing attack.*

3. 5-Stage Automated Ingestion Radar

What It Does: A visual live progress radar that shows the engine executing 5 sequential checks: RFC-5322 extraction, cryptographic verification, urgency NLP, hop geolocation, and custody seal synthesis.

How to Use: *When you click 'Analyze Now', you watch green checkmarks light up step-by-step in under 500 milliseconds.*

4. Radial Threat Score Gauge (0–100 Scale)

What It Does: A color-coded speedometer gauge that synthesizes all technical checks into an intuitive security score: 0–29 is SAFE (Green), 30–69 is SUSPICIOUS (Amber), and 70–100 is MALICIOUS (Red).

Example: *An email with failed SPF (+25), failed DKIM (+20), failed DMARC (+25), and urgent threats (+20) receives a Threat Score of 90/100 (MALICIOUS).*

5. Interactive Leaflet World Map (MTA Hop Tracer)

What It Does: A vector world map powered by Leaflet.js that reverse-unrolls the server hops and draws flight polyline curves connecting each transit city across the globe.

Example: *A red pulsing dot appears in Moscow, Russia (Hop #1 Origin), connecting via blue flight line to Amsterdam, Netherlands (Hop #2 Relay), revealing the true path.*

6. Foreign Origin Mismatch Flag

What It Does: An automated discrepancy detector that triggers when the claimed organization's country differs from the actual physical server origin IP.

Example: *An email claims to be 'State Bank of India Corporate Support', but the server IP is physically located in St. Petersburg, Russia. The system triggers a bright red 'Origin Mismatch' badge.*

7. Conversational Forensic AI Copilot

What It Does: An intelligent chatbot widget (bottom-right corner) that translates dense technical headers into plain English. You can ask it any question about the case or email security.

Example Questions: *'Why was this email flagged?', 'What is SPF?', 'Is it safe to click the link?', 'Where was this email sent from?'*

8. Court-Admissible PDF Forensic Report

What It Does: Programmatically generates a formal, tamper-evident cyber incident audit report with case details, hop tables, and the SHA-256 custody seal, ready for legal filing.

How to Use: Click 'Forensic Report (PDF)' on any scan result page to download a sealed document ready for cyber crime police submission.

9. 1-Click CERT-In Incident Dispatch

What It Does: Formats extracted attacker IPs, spoofed domains, and payload hashes into the standardized reporting format mandated by the Indian Computer Emergency Response Team within the 6-hour window.

How to Use: Click 'Report to CERT-In' to generate an official incident reference ticket (e.g. CERT-IN-2026-CASE-98F2).

10. Zero-Downtime Dual-Engine Architecture

What It Does: Combines a cloud Python FastAPI backend with an instant in-browser client fallback. If the cloud server is sleeping or slow, the browser parses the email locally in <50ms.

Benefit: The website NEVER crashes, freezes, or shows 'Network Error' during presentations or poor Wi-Fi.

11. 1-Click Desktop Launcher (START_MAILSHIELD.bat)

What It Does: A Windows desktop shortcut that boots both the Python FastAPI server and Vite frontend locally on localhost without needing internet.

How to Use: Double-click START_MAILSHIELD.bat on your Desktop to run offline during emergency network outages.